Rubber-Band Dynamics Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Valence Gap and Re-alignment Analysis

PENPAL Project

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Contents

1 Overview

This notebook compares **rubber-band dynamics** — the tendency for divergent valence to "snap back" toward alignment — across all three experimental conditions:

- **Human-AI**: Human co-writing with an LLM
- **Human-Human**: Two humans co-writing together
- **AI-AI**: Two AI agents co-writing (simulated baseline)

The Rubber-Band Hypothesis: When there is a valence gap (emotional divergence) between co-authors, the subsequent turn should show a correction toward alignment. This is operationalized as:

- **Valence Gap**: Difference between Author 1 and Author 2 valence at turn t
- **Gap Change**: Change in valence gap from turn t to turn t+1
- **Rubber-Band Effect**: Negative correlation between gap and gap change (divergence predicts re-convergence)

Key Questions: 1. Is the rubber-band effect present in all conditions? 2. Is the effect stronger in Human-AI (AI accommodating) vs Human-Human (mutual accommodation)? 3. Does AI-AI show the strongest correction (most "compliant" dynamic)? 4. How does the correction direction differ across conditions? 5. Could the rubber-band effect be an inherent mathematical phenomenon, rather than behavioural/psycholinguistical?

2 Load Data

```
## Loaded conditions: human-ai, human-human, ai-ai
```

```
## # A tibble: 3 x 3
##   condition   n_stories n_turns
##   <fct>           <int>   <int>
## 1 Human-AI           87     870
## 2 Human-Human        38     380
## 3 AI-AI              40     400
```

3 Preprocessing: Valence Gap Computation

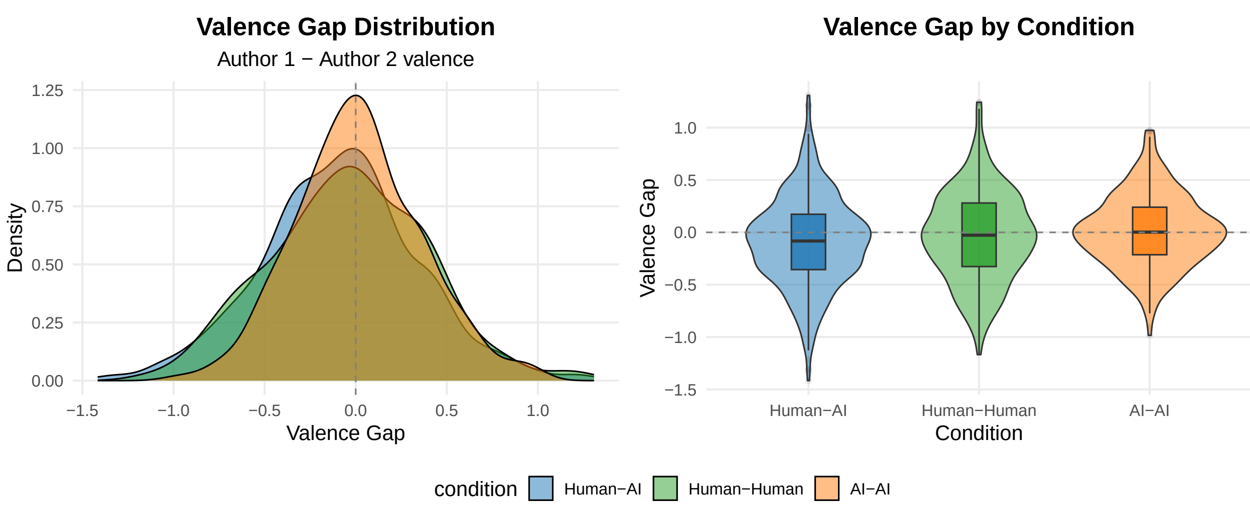
```
## Analysis dataset: 1485 turns across 165 stories
```

4 Descriptive Statistics

Table 1: Valence gap statistics by condition

Condition	N Turns	Mean Gap	SD Gap	Mean Δ Gap	SD Δ Gap	Rubber-Band r
Human-AI	783	-0.087	0.428	0.009	0.572	-0.666
Human-Human	342	-0.038	0.431	-0.012	0.620	-0.720
AI-AI	360	0.014	0.345	-0.006	0.465	-0.673

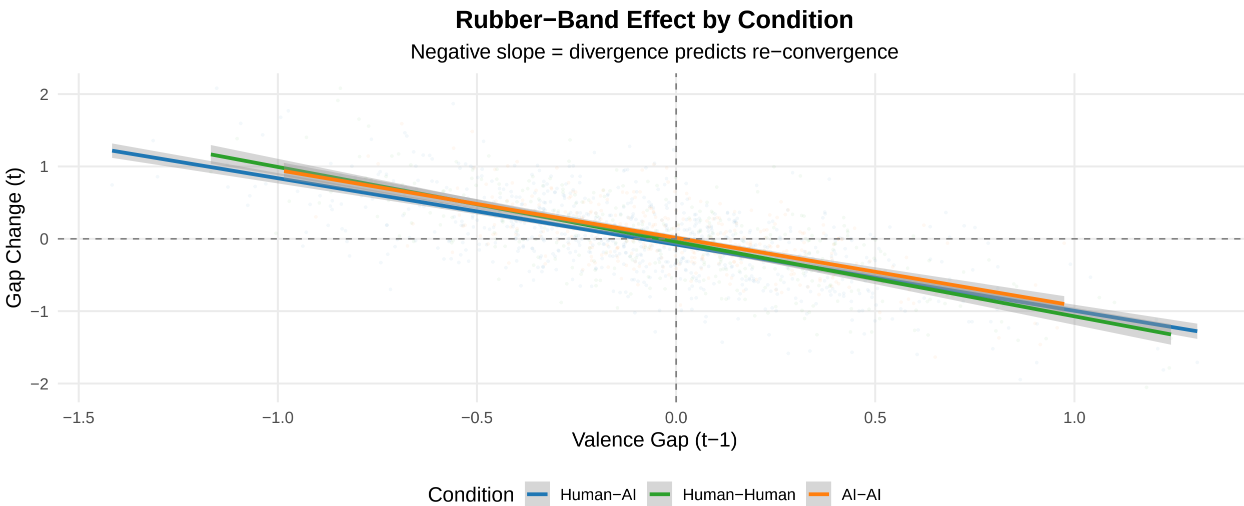
5 Valence Gap Distribution



6 Rubber-Band Effect Analysis

6.1 Core Test: Gap → Gap Change

The rubber-band hypothesis predicts that **previous gap negatively predicts gap change** — when there's divergence, the system corrects toward alignment.



6.2 Within-Condition Models

Table 2: Rubber-band effect by condition (Gap_t-1 → ΔGap_t)

condition	estimate	se	ci_lower	ci_upper	t	p	sig	interpretation
Human-AI	-1.022	0.037	-1.095	-0.950	-27.727	0	***	Rubber-band present
Human-Human	-1.037	0.054	-1.143	-0.931	-19.211	0	***	Rubber-band present
AI-AI	-0.938	0.054	-1.044	-0.831	-17.221	0	***	Rubber-band present

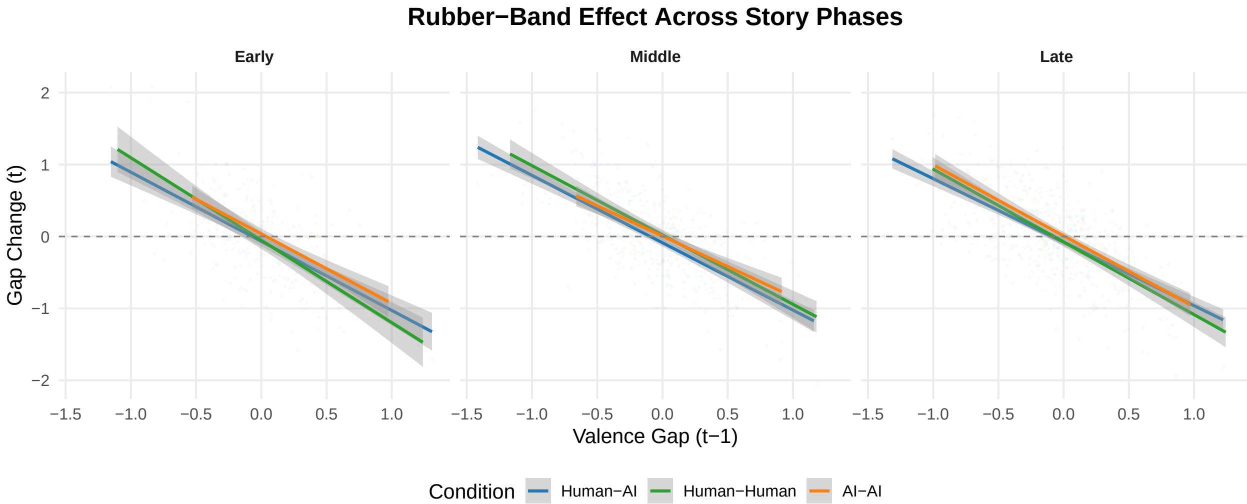
6.3 Cross-Condition Comparison

```
## === Cross-Condition Rubber-Band Model ===

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: gap_change ~ valence_gap_lag1 * condition + (1 | conversation_id)
##    Data: df_analysis
##
## REML criterion at convergence: 1576.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.1062 -0.6280 -0.0063  0.6046  3.5788
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.009746 0.09872
##  Residual                    0.158658 0.39832
## Number of obs: 1485, groups:  conversation_id, 165
```

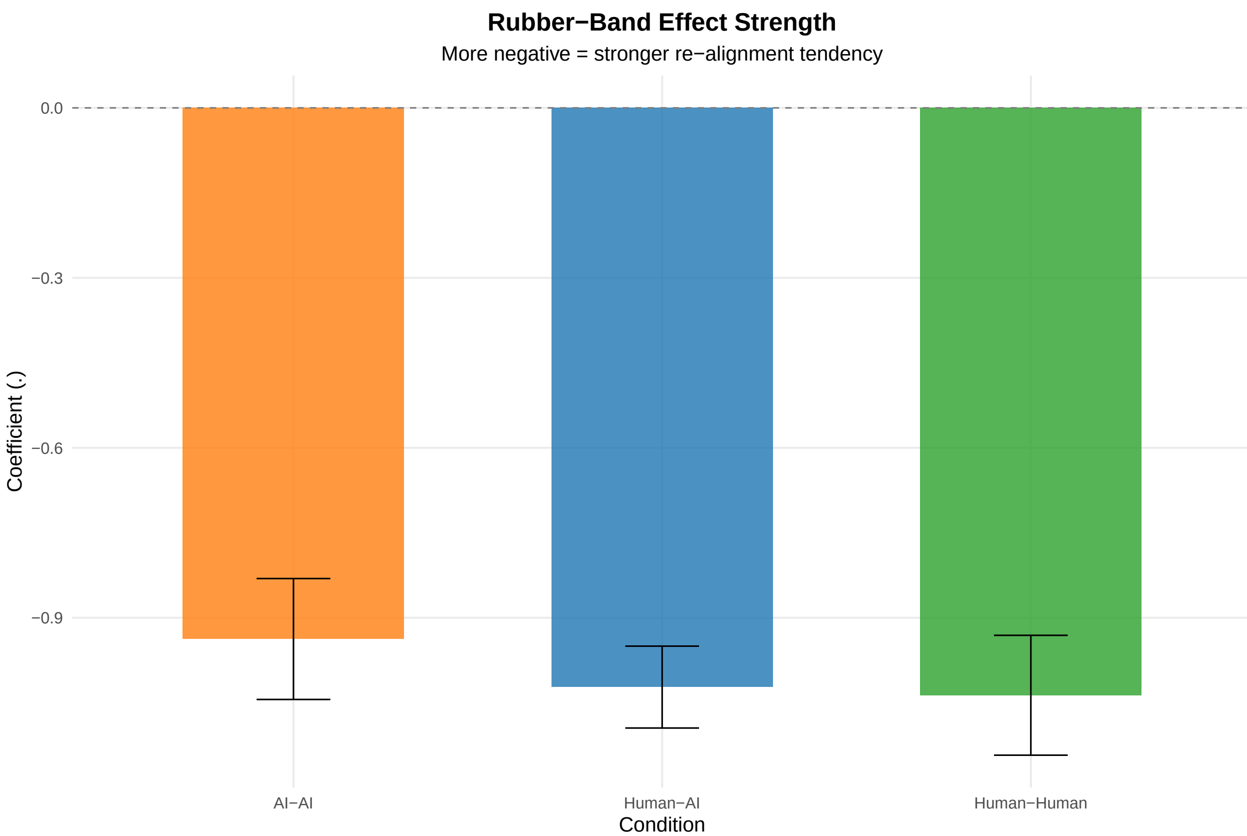
```
##
## Fixed effects:
##                                         Estimate Std. Error         df t value
## (Intercept)                             -0.08562    0.01806  133.97215  -4.740
## valence_gap_lag1                        -0.98567    0.03556 1460.67632 -27.716
## conditionHuman-Human                     0.04530    0.03238  128.61970   1.399
## conditionAI-AI                           0.09944    0.03182  128.70945   3.125
## valence_gap_lag1:conditionHuman-Human   -0.08459    0.06208 1475.69768  -1.363
## valence_gap_lag1:conditionAI-AI          0.01175    0.07351 1474.15996   0.160
##                                       Pr(>|t|)
## (Intercept)                           5.38e-06 ***
## valence_gap_lag1                       < 2e-16 ***
## conditionHuman-Human                    0.1642
## conditionAI-AI                          0.0022 **
## valence_gap_lag1:conditionHuman-Human   0.1732
## valence_gap_lag1:conditionAI-AI         0.8730
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) vln__1 cndH-H cAI-AI v__1:H
## vlnc_gp_lg1  0.188
## cndtnHmn-Hm -0.558 -0.105
## condtnAI-AI -0.568 -0.107  0.317
## vlnc__1:H-H -0.108 -0.573  0.094  0.061
## vl__1:AI-AI -0.091 -0.484  0.051  0.016  0.277

##
## === Rubber-Band Slopes by Condition ===

##  condition   valence_gap_lag1.trend     SE   df lower.CL upper.CL
##  Human-AI                    -0.986 0.0356 1465    -1.06   -0.916
##  Human-Human                 -1.070 0.0509 1462    -1.17   -0.970
##  AI-AI                       -0.974 0.0644 1466    -1.10   -0.848
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

##
## === Pairwise Slope Comparisons ===

##  contrast                    estimate     SE   df t.ratio p.value
##  (Human-AI) - (Human-Human)    0.0846 0.0622 1476   1.361  0.5212
##  (Human-AI) - (AI-AI)         -0.0117 0.0736 1475  -0.160  1.0000
##  (Human-Human) - (AI-AI)      -0.0963 0.0821 1464  -1.173  0.7226
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
```

7 Temporal Dynamics

How does the rubber-band effect evolve across the story?



8 Summary Visualization



9 Key Findings

Based on the analysis above:

1. **Rubber-band presence**: [Interpret whether the effect is present in all conditions]
2. **Effect strength comparison**: [Compare magnitude across Human-AI, Human-Human, AI-AI]

3. **Temporal patterns**: [Note if the effect changes across story phases]
4. **AI-AI baseline**: [Compare simulated baseline to empirical conditions]

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